

PROJECT REPORT

AI Customer Support Ticket Triage

Intelligent Ticket Classification, Priority Detection
& Automated Support Routing U

Submitted by

SHIFANA SHERIN J

Department of Artificial Intelligence and Data Science

K .RAMAKRISHNAN COLLEGE OF ENGINEERING

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1. Project Title

AI Customer Support Ticket Triage

Subtitle: Intelligent Ticket Classification, Priority Detection & Automated Support Routing

2. Abstract

- Customer support teams receive a large number of tickets every day through email, websites, mobile applications, and other communication channels. Manually reading, categorizing, prioritizing, and assigning these tickets can be time-consuming and may result in delayed responses or incorrect routing.
- The AI Customer Support Ticket Triage system is an Artificial Intelligence and Natural Language Processing based solution that automatically analyzes incoming customer support tickets. The system identifies the ticket category, urgency, sentiment, priority, and appropriate support team based on the content of the customer's request.
- The system uses NLP and machine learning techniques to understand ticket descriptions and classify them into categories such as technical issues, billing problems, account-related issues, product complaints, and general queries. It can also detect urgent or negative customer messages and assign suitable priority levels. The processed tickets are displayed through an interactive dashboard, allowing support teams to monitor workload, track ticket status, and handle high-priority issues efficiently.

3. Problem Statement

- Customer support departments receive numerous tickets containing different types of problems, requests, complaints, and inquiries. Manually reviewing each ticket to determine its category, priority, sentiment, and responsible support team requires considerable time and may lead to inconsistent classification.
- Low-priority and high-priority tickets may be mixed together, making it difficult for support teams to identify urgent customer issues quickly. Incorrect routing can also cause delays because tickets may be assigned to teams that are not responsible for resolving them.
- Therefore, there is a need for an AI-based ticket triage system that can automatically understand customer messages, classify tickets, determine their priority and sentiment, and route them to the appropriate support team. This can reduce manual effort, improve response time, and provide better customer support management.

4. Objectives

The main objectives of this project are:

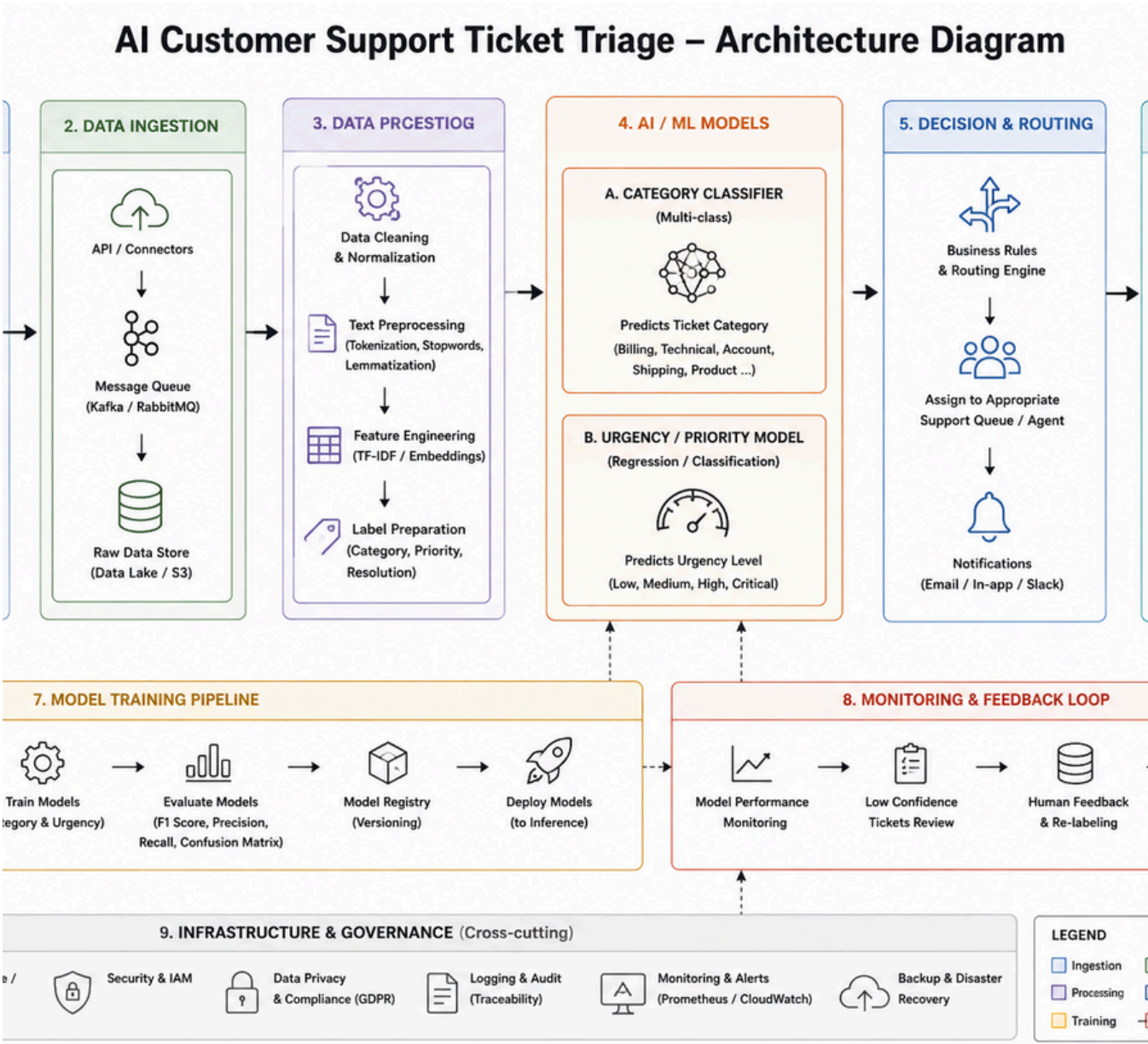
- To automatically analyze incoming customer support tickets.
- To classify tickets into appropriate categories.
- To identify the urgency and priority of each ticket.
- To detect customer sentiment such as positive, neutral, or negative.
- To automatically assign tickets to suitable support teams.
- To identify critical or urgent customer complaints.
- To reduce manual ticket classification and routing.
- To provide confidence scores for AI predictions..

- To detect duplicate or similar support tickets.
- To identify frequently occurring customer issues.
- To provide an interactive dashboard for ticket monitoring.
- To track ticket status and resolution progress.
- To improve response time and support-team efficiency.
- To allow human agents to review and correct uncertain classifications.

5. Proposed System

- The proposed system is an AI-powered customer support ticket triage platform that automatically processes and organizes incoming support requests.
- When a customer submits a ticket, the system first performs text preprocessing to remove unnecessary content and prepare the message for analysis. NLP techniques are then used to understand the ticket and extract important information.
- A machine learning model classifies the ticket according to its category, priority, urgency, and sentiment. Based on the classification results, the system recommends or automatically assigns the ticket to the appropriate support team.
- Tickets with high urgency or negative sentiment can be highlighted for immediate attention. Low-confidence predictions can be sent to human agents for verification. Duplicate and similar tickets can also be identified to avoid repeated work.
- The final results are displayed through an interactive dashboard containing ticket categories, priority levels, sentiment, assigned teams, ticket status, and workload analytics.
- System Workflow
- Customer Ticket → Text Preprocessing → NLP Analysis → Feature Extraction → AI/ML Classification → Category + Priority + Sentiment Detection → Confidence Scoring → Team Assignment → Database → Dashboard & Monitoring

6. System Architecture



7. Modules Used

1. User Authentication Module

Provides secure login and access control for support agents and administrators.

2. Ticket Input Module

Collects customer tickets from web forms, email, chat, or other supported sources.

3. Text Preprocessing Module

Cleans customer messages by removing unwanted symbols, formatting, and irrelevant text.

4. NLP Processing Module

Analyzes the language and context of customer messages.

5. Feature Extraction Module

Converts ticket text into machine-readable features using TF-IDF, embeddings, or transformer-based representations.

6. Ticket Classification Module

Classifies tickets into categories such as:

Technical Issue

Billing

Account

Product Issue

Complaint

General Query

7. Priority & Urgency Detection Module

Determines whether a ticket is low, medium, high, or critical priority.

8. Sentiment Analysis Module

Identifies customer sentiment such as positive, neutral, or negative.

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9. Smart Routing Module

Assigns tickets to the most suitable support team or department.

10. Duplicate Detection Module

Identifies similar or repeated tickets to reduce duplicate handling.

11. Confidence & Validation Module

Provides confidence scores and flags uncertain predictions for human review.

12. Dashboard & Analytics Module

Displays ticket statistics, workload, priority distribution, sentiment, and team performance.

13. Ticket Tracking Module

Maintains ticket status such as Open, In Progress, Pending, Resolved, and Closed.

14. Database Module

Stores customer tickets, classifications, assignments, status, and historical information

8. Technologies Used

Technology	Purpose
Python	Main programming language
Natural Language Processing	Understanding customer ticket text
Machine Learning	Ticket classification and priority prediction
NLTK / spaCy	Text preprocessing and linguistic analysis
Scikit-learn	ML models and feature processing
TF-IDF	Text feature extraction
BERT / Transformers	Advanced contextual understanding
Pandas	Data processing and analysis
SQLite / MySQL	Ticket and user data storage
Streamlit	Interactive web dashboard
Plotly / Matplotlib	Charts and analytics
Git & GitHub	Version control

9.Key Features

- Automatic Ticket Classification – Categorizes incoming tickets based on their issue type.
- Priority Detection – Assigns Low, Medium, High, or Critical priority.
- Urgency Detection – Identifies tickets that require immediate attention.
- Sentiment Analysis – Detects customer emotions such as positive, neutral, and negative.
- Smart Ticket Routing – Automatically directs tickets to the appropriate support team.
- AI-Based Text Analysis – Uses NLP to understand the customer's problem and intent.
- Keyword Extraction – Identifies important words and phrases from customer messages.
- Intent Detection – Determines what the customer is requesting or reporting.
- Confidence Scoring – Provides a confidence score for AI predictions.
- Low-Confidence Flagging – Sends uncertain tickets for human verification.
- Duplicate Ticket Detection – Identifies repeated or similar customer complaints.
- Similar Ticket Search – Finds previously handled tickets with similar problems.
- Automatic Ticket Tagging – Adds relevant tags such as Billing, Technical, Account, or Complaint.
- SLA Monitoring – Tracks response and resolution deadlines.
- Escalation Detection – Identifies tickets that may require escalation to senior support teams.
- Ticket Status Tracking – Tracks tickets as Open, In Progress, Pending, Resolved, or Closed.

10. Advantages

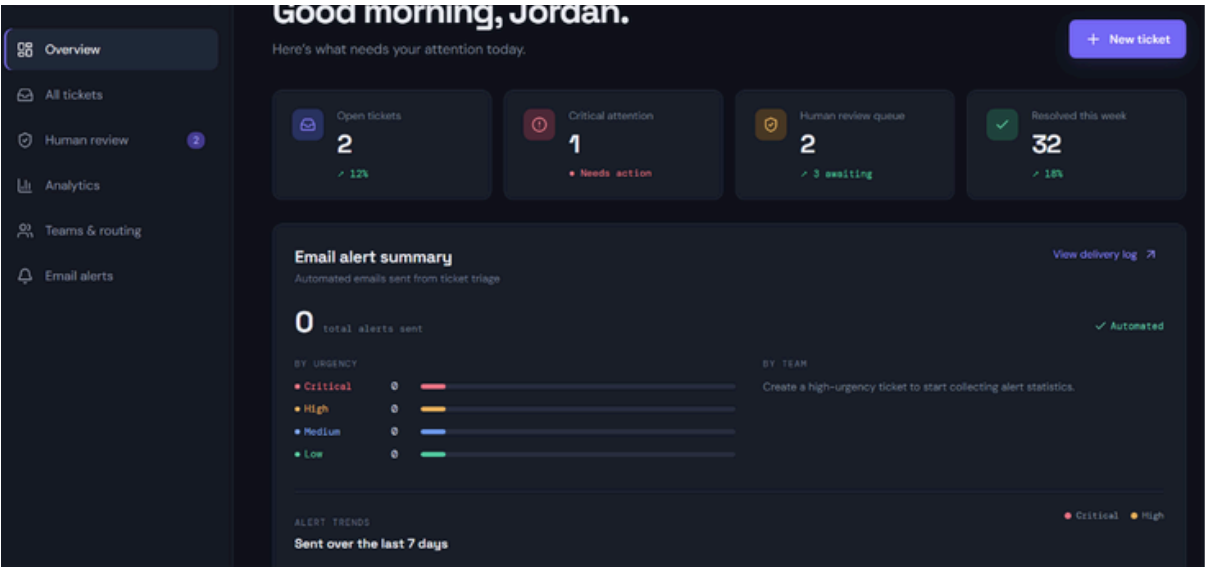
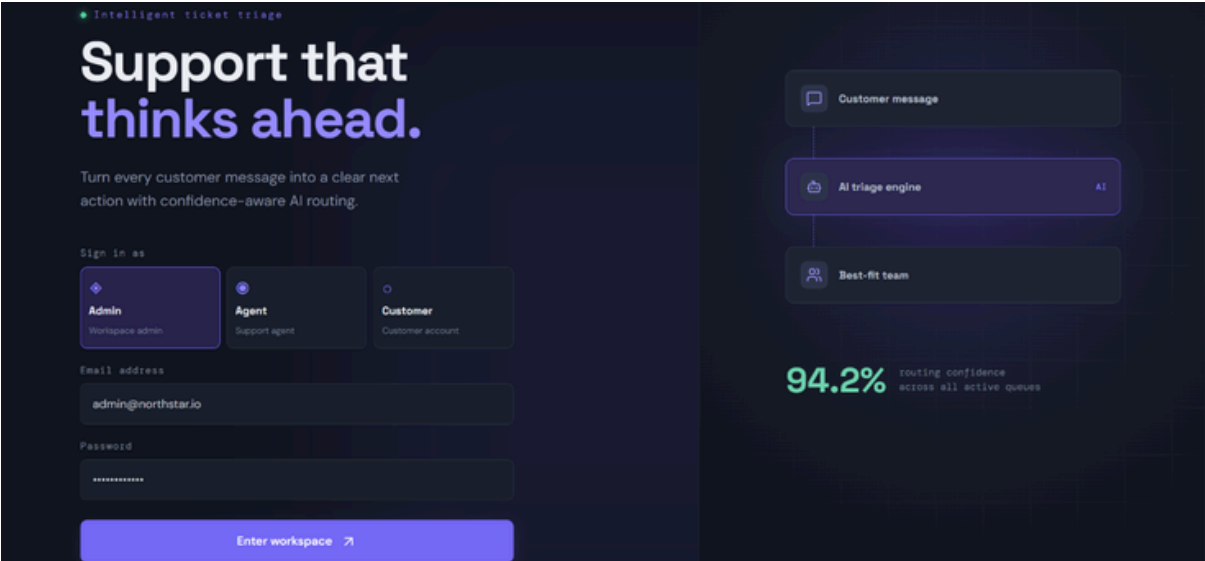
- 1.Reduces Manual Work – Automatically categorizes incoming support tickets.
- 2.Faster Ticket Processing – Reduces the time required to review each ticket.
- 3.Improves Ticket Routing – Sends tickets to the appropriate support team.
- 4.Priority-Based Handling – Helps agents focus on urgent and critical issues first.
- 5.Sentiment Detection – Identifies frustrated or dissatisfied customers.
- 6.Improves Response Time – Helps teams respond to important issues faster.
- 7.Consistent Classification – Applies the same classification criteria to tickets.
- 8.Duplicate Detection – Reduces repeated handling of similar tickets.
- 9.Workload Visibility – Dashboard shows the workload of different support teams.
- 10.Scalable – Can process a large number of customer tickets.
- 11.Better Customer Experience – Faster and more accurate ticket handling can improve customer satisfaction.
- 12.Human Verification – Allows agents to correct uncertain AI predictions.

11. Limitations

- Language Ambiguity – AI may misunderstand unclear or informal customer messages.
- Training Data Dependency – Accuracy depends on the quality and quantity of training data.
- Incorrect Classification – Some complex tickets may be placed in the wrong category.

- Sentiment Errors – Sarcasm and indirect language can be difficult to detect.
- Priority Prediction Errors – AI may not always correctly understand the urgency of a request.
- Limited Domain Knowledge – The system may require domain-specific training for specialized businesses.
- Human Review Required – Critical or low-confidence tickets may still need manual verification.
- Privacy Concerns – Customer tickets may contain personal or confidential information.
- Integration Complexity – Connecting the system with existing customer-support platforms may require additional development.
- Model Maintenance – The AI model may need regular retraining as customer issues and terminology change.

12.Output Screenshot



NEW INBOUND SIGNAL



Create support ticket

Customer name

Aarav Mehta

Subject

What does the customer need help with?

Description

Add the full customer message...



AI will predict category, urgency, confidence, and the best-fit support team automatically.

Cancel

Run AI triage

Northstar Cloud

Overview

All tickets

Human review

Analytics

Teams & routing

Email alerts

Settings

PERFORMANCE CENTER

Analytics

A live view of routing quality, workload, and customer outcomes.

Export report

Ticket volume

Last 7 days - all channels

18.4%



By category

Current ticket distribution



Avg. first response
18m

> 6m faster

Resolution rate
86.4%

> 4.2%

AI corrections
8.7%

> Within target

SLA at risk
12

Requires attention

WORKSPACE
Northstar Cloud

Overview

All tickets

Human review 2

Analytics

Teams & routing

Email alerts

Settings

WORKSPACE CONFIGURATION

Teams & routing

Configure where AI sends each customer signal.

+ Add team

Billing Operations

Billing · Refunds · Payments

12 agents 82% load

Technical Support

Bugs · API · Integrations

18 agents 68% load

Account Security

Login · Access · Security

8 agents 91% load

Customer Success

Plans · Seats · Onboarding

14 agents 54% load

Routing rules

Edit rules

Rules are evaluated top to bottom

1	Critical + Technical Support	Priority Technical	Always
2	Security / Login	Account Security	High confidence
3	Billing / Refund	Billing Operations	Always
4	Everything else	General Support	Fallback

WORKSPACE
Northstar Cloud

Overview

All tickets

Human review 2

Analytics

Teams & routing

Email alerts

Settings

AUTOMATED NOTIFICATIONS

Email alerts

High-urgency tickets automatically notify the assigned support team.

Automation active

High and Critical ticket alerts

When AI identifies a high-urgency ticket, SignalDesk creates an email notification for the team on-call address. This demo records the email safely in the notification center; connect SMTP or SendGrid credentials to deliver externally.

Delivery log

Export CSV

Delivery simulation healthy

0 of 0 automated emails shown

Filter urgency All Critical High

No matching alerts

Create a High or Critical ticket to trigger an email alert.

13. Future Scope

- Real-Time Ticket Triage – Classify and route tickets immediately when they are received.
- Multilingual Support – Process customer requests in multiple languages.
- Voice Ticket Processing – Convert customer calls or voice messages into text and analyze them.
- Advanced Generative AI – Generate suitable responses for common customer queries.
- Automatic Response Generation – Provide suggested replies to support agents.
- Predictive Ticket Escalation – Predict which tickets are likely to require escalation.
- Smart SLA Monitoring – Monitor service-level deadlines and warn agents about possible SLA violations.
- Advanced Customer Sentiment Analysis – Track changes in customer sentiment during conversations.
- Omnichannel Integration – Integrate email, live chat, social media, mobile applications, and call-center systems.
- Continuous Learning – Use corrected classifications and agent feedback to improve the AI model.
- Enterprise Integration – Connect with CRM and customer-support platforms.
- AI Support Assistant – Develop an intelligent assistant that helps agents resolve tickets more quickly.

14 . Conclusion

- The AI Customer Support Ticket Triage system provides an intelligent approach for automatically analyzing, classifying, prioritizing, and routing customer support tickets. By combining Artificial Intelligence, Natural Language Processing, and Machine Learning, the system can understand customer messages and identify important information such as ticket category, priority, urgency, and sentiment.
- The system reduces the manual effort involved in ticket management and helps support teams identify critical issues quickly. Its smart routing, duplicate detection, confidence scoring, and interactive dashboard features improve the efficiency of customer-support operations.
- Overall, the proposed system can help organizations reduce response time, improve ticket management, distribute workload effectively, and provide a better customer experience. Future enhancements such as multilingual processing, voice analysis, generative AI, SLA prediction, and CRM integration can further improve its capabilities.

15 REFERENCE

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Python Documentation

Scikit-learn Documentation

spaCy Documentation

Hugging Face Transformers Documentation

Streamlit Documentation

